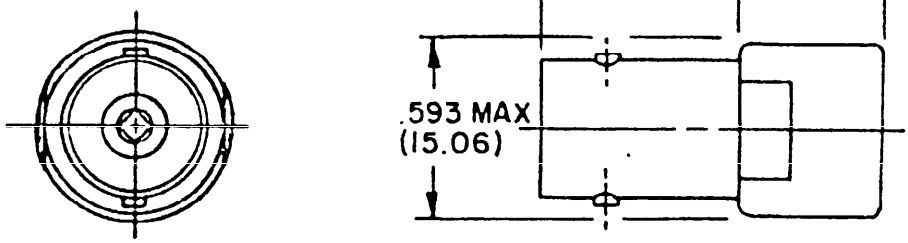
MILITARY SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY, (SERIES MHV (CABLED) - PLUG, SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-39012.

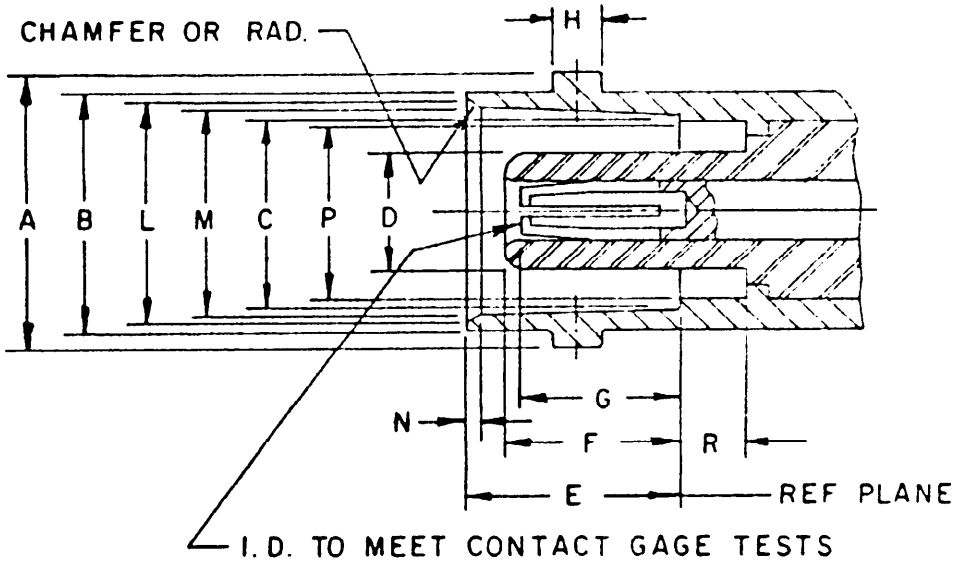
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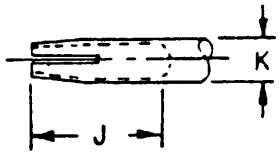


NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. For dimension A, see table I.
4. Dimension .593 is the largest overall diameter of the connector.
5. Wrench flats are to accommodate standard wrench openings per H-28, app. 10.
6. All undimensioned pictorial representations are for reference purposes only.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.

FIGURE 1. General configuration.

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LTR	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
A	.432 (10.97)	.436 (11.07)
B	.378 (9.60)	.382 (9.70)
C	.319(8.10)	.321 (8.15)
D		.186 (4.72)
E	.327 (8.31)	.335 (8.51)
F	.289 (7.34)	.311 (7.90)
G	.253 (6.43)	.280 (7.11)
H	.075 (1.91)	.081 (2.06)
J	.270 (6.86)	
K	.081 (2.06)	.091 (2.31)
L	.346 (8.79)	.356 (9.04)
M	.327 (8.31)	.333 (8.46)
N	.015 (.38)	.030 (.76)
P	.284 (7.21)	.290 (7.37)
R	.086 (2.18)	

FIGURE 2. Mating dimensions for socket terminations.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: Not applicable.

Voltage rating:
1600 volts rms, maximum working voltage at sea level.
375 volts rms, maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - 3 pounds, maximum.
Torque - 2-1/2 inch-pounds, maximum.

Mating characteristics:
See figure 2 for dimensions.
Outer contact:
Test ring ID - .319 maximum, 16 microinch finish.
Insertion force - 5 pounds, maximum when inserted a minimum of .093.
Contacts with slotted members: Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms, minimum.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact	2.0	2.5
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 5,000 volts rms, minimum at sea level.

Corona level:
Voltage - 1245 volts rms.
Altitude - 70,000 feet.

RF high potential withstanding voltage: Not applicable.

Connector durability:
500 cycles minimum at 12 cycles/min maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.
Initial - 5 pounds, maximum.
Final - 5 pounds maximum; 1 pound minimum.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202. No discontinuities.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition I.
Acceleration - 50 G's at 7 milliseconds. No discontinuities.

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TABLE I. Dash number, cross reference and dimensions.

Dash No.	Applicable cable number	Dim	Inches-millimeters maximum	Inches-millimeters minimum
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)				
0101	M17/29-RG59 M17/30-RG62 M17/90-RG71 M17/97-RG210	A	1.500 (38.1)	
		B	.875 (22.225)	.850 (21.44)

Thermal shock: Method 107, test condition B, MIL-STD-202, ex , c test high temperature shall be +85°C. High temperature shall be +2' °C for connectors using +200°C cable (see table I).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Cable retention force:
Noncrimp assemblies - 40 pounds, minimum.

Coupling mechanism retention force: Not applicable.

Part number: M39012/101-0001

TABLE II. Supersession and cross reference.

Part number	Superseded numbers			
	FSCM 74868	FSCM 91836	FSCM 77820	Type designation
M39012/101-0001	28250		3064-1	UG1016A

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy -
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AM, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3065-2)

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